

Natural Hydrogen Exploration in the Bristol Channel–Bray Fault System: Integrating Image–Pyramid Instance Segmentation for SCD Detection with Soil Gas Field Measurements

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Abstract

Sub-circular depressions (SCDs) are commonly associated with natural hydrogen (H_2) production and active H_2 migration. The SCDs in Southwest England are currently undetectable by current SCD detection models due to their small size and spectral characteristics. This study develops a new deep-learning model capable of detecting SCDs irrespective of their size or location. We subsequently deploy the model in Southwest England to identify SCD clusters, and perform a soil gas sampling campaign, providing the first investigation into natural H_2 prospectivity in the United Kingdom. We utilise a Mask R-CNN architecture using topographic data to mitigate the inconsistent spectral characteristics of SCDs. The model is trained using scale-normalised samples, and an image-pyramid is utilised at inference to maximise the receptive field of the model. The model achieves an F1 score of 0.82 in validation, and an average F1 score of 0.70 in region-level leave-one-out cross-validation. At inference, the model is capable of detecting SCDs ranging from 20 to 1000 metres in diameter, in completely unseen regions. This important contribution enables automatic cross-geography detection of SCD at a wider range of scales. In Southwest England, 2752 SCDs are identified across 11460 km^2 , including the identification of a cluster near the Bristol Channel Fault Zone, Somerset. Here, SCDs are strongly spatially associated to mapped faults ($D^+=0.221$), especially the major Variscan strike-slip Watchet-Cothelstone-Hatch fault ($D^+=0.184$). Soil gas H_2 concentrations of up to 300 ppm indicate the presence of an active H_2 migration system. The results demonstrate that the Bristol Channel-Bray Fault system has significant potential for H_2 exploration in the United Kingdom, possibly correlating with well-documented H_2 discoveries near the Bray Fault, France.



1 Introduction

Natural hydrogen (H_2) has been the subject of significant attention for transformative potential in the energy transition (Ellis and Gelman, 2024). While H_2 is already widely used in the energy sector, it is primarily produced using expensive or carbon-intensive processes, such as steam-methane reforming or electrolysis (IEA, 2025). Conversely, natural H_2 is produced through subsurface geological such as serpentinisation or radiolysis (Torghabeh et al., 2026). Only 1×10^5 Mt of this naturally produced H_2 would provide a completely carbon-free resource yielding more than double the amount of energy in all known natural gas reserves on Earth (Ellis and Gelman, 2024). Current H_2 exploration techniques rely on near-surface soil gas surveys and borehole data (Torghabeh et al., 2026), but recent discoveries of surface H_2 leaks associated with sub-circular depressions (SCDs) (Frery et al., 2021; Larin et al., 2015; Prinzhofer et al., 2019; Zgonnik et al., 2015) have initiated an interest in deep learning based exploration strategies (Ginzburg et al., 2025; Roche et al., 2026). This study aims to develop and deploy a generalised instance segmentation model, utilising image-pyramid inference strategy to identify clusters of SCDs in Southwest England. We subsequently report results from a soil gas sampling campaign in Bristol Channel-Bray Fault zone, Somerset, providing the first investigation into H_2 prospectivity in the United Kingdom.

The Bristol Channel-Bray Fault is a major strike-slip Variscan structure running from Southwest England to the Paris Basin, France, responsible for up to 100 kilometres of dextral displacement across Northwest Europe in Carboniferous times (Shail and Leveridge, 2009). In Southwest England, the Bristol Channel-Bray Fault bounds the Rhenohercynian Zone to the north and east, juxtaposing Devonian rocks from Southwest England against the Ordovician age Avalonian terrane in Wales and central England. Several SCDs have been previously found along this fault in the Paris Basin. Hydrogen concentrations along the Bray Fault are variable but often considerably elevated, with subsurface concentrations reaching as high as 52 vol% in the Dogger Aquifer (Lefeuvre et al., 2024), and a maximum 670 ppm measured within the soil (Léger et al., 2025). While interest in hydrogen exploration is growing globally, no investigations have been made into the resource’s potential in the United Kingdom, despite significant discoveries in geologically-linked locations in the Paris Basin, France.

SCDs are promising indicators of subsurface H_2 production and migration, and may be linked with large subsurface accumulations of H_2 . The mechanism for SCD formation is poorly understood, but they have been proposed to form from mechanical compaction due to gas inflow and two-phase subsurface pore-pressure interactions (Schopfer et al., 2025). Despite being a nascent field of research, recent studies (Ginzburg et al., 2025; Roche et al., 2026) have successfully employed semantic segmentation utilising satellite imagery to detect SCDs at scale. However, these approaches struggle to generalise across terrain types and land uses, as SCDs can exhibit distinct optical surface expressions depending on climatic and agricultural conditions. SCDs can also range in size from 20 metres (as described in this study) to 2 kilometres in diameter (Frery et al., 2021; Larin et al., 2015; Prinzhofer et al., 2019; Schopfer et al., 2025), and current models only utilise at one resolution at inference, limiting the receptive field of the model.

Here, we initially utilise a localised region-specific exploratory model to detect SCD clusters surrounding Variscan structures related to the Bristol-Bray system in the United Kingdom. The study develops a generalised cross-geography model and inference strategy capable of identifying SCDs at any scale in completely unseen regions, which can reproduce the predictions of the localised region-specific model in the UK. Model performance is evaluated in the context of SCD exploration through validation in regions with previously documented SCDs and H_2 migration, such as the São Francisco Basin, Brazil and Moora, Western Australia. Finally, we report results from a near-surface geochemical soil gas sampling campaign at predicted SCD clusters in Somerset, UK, and provide the first investigation into H_2 prospectivity within the United Kingdom.

2 Methods

2.1 Data Preparation and Network Architecture

We utilise Digital Elevation Model (DEM) data to capture the morphological characteristics of SCDs without relying on geographically varying spectral signatures. The Copernicus DEM (30 metre resolution) is used as the primary data source. Where available, higher resolution national datasets are used (UK, DEFRA, 2022; France, IGN, 2021; USA, USGS, 2020). Orthorectified optical satellite data at 3 metre resolution is obtained from Planet Labs (Planet, 2026).

Five channels are input to the model: DEM, Slope, Laplace (curvature), Residual Relief, and Multidirectional Hillshade (Figure 1a). Residual Relief (RR), a parameter to capture in-slope topography, is calculated by applying a Gaussian smoothing to the DEM and negating the difference from the raw DEM. Each channel is normalised at the tile level. The chosen model architecture (Fig. 1b) is based on the Mask R-CNN (He et al., 2018). This architecture is widely used in medical imaging for instance segmentation and object detection (Chiwahne et al., 2024). To adapt the Mask R-CNN for SCD detection, the anchor sizes are reduced and the elongation is decreased to align with the typical dimensions and circular geometry of target SCDs.

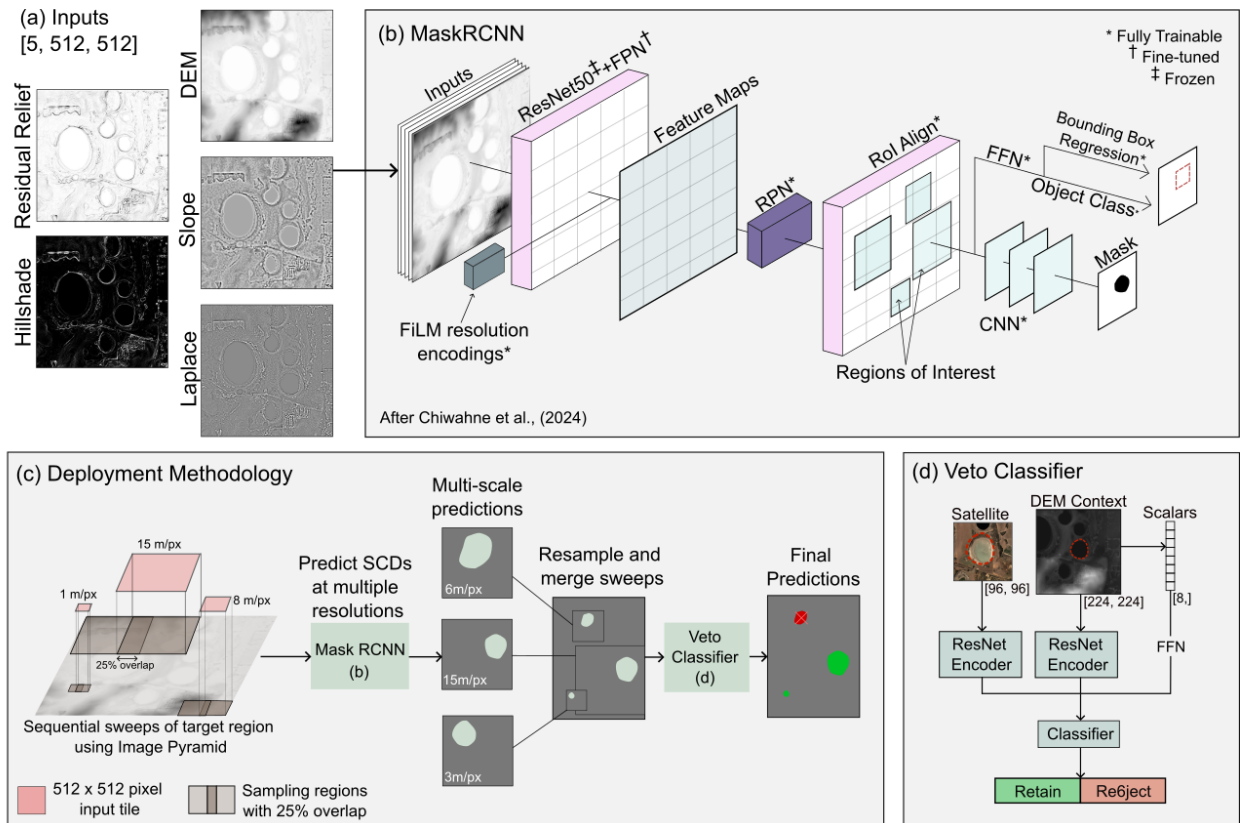


Figure 1: Schematic of deep learning methodology for scale-invariant SCD detection: (a) adapted Mask R-CNN architecture for SCD detection, (b) inference strategy using a Scale-Normalised Image Pyramid (SNIP) to maximise the receptive field of the inference mechanism, (c) DEM and DEM-derivative input channels, (d) architecture of the veto classifier used to screen candidates. RPN = Regional Proposal Network, FPN = Feature Pyramid Network, ROI = Region of Interest, FiLM = Feature-Wise Linear Modulation, CNN = Convolutional Neural Network, FFN = Feed Forward Network.

The training dataset contains 284 positive and 306 negative tiles. Negative tiles contain no SCD instances and are sampled from a large range of geographies. Positive tiles contain at least one SCD instance. Training samples are generated from different regions with previously identified SCDs: São Francisco Basin, Brazil (-16.452391N, -45.248511E; Prinzhofer et al., 2019), Borisoglebsk, Russia (51.135297N, 41.911169E; Larin et al., 2015), Moora, Australia (-30.619787N, 115.968694E; Frery et al., 2021), Stiles, USA (31.472409N, -101.597082E), Coubert, France (48.672204N, 2.756956E). The training regions contain SCDs ranging in physical size and morphology to maximise the distribution of characteristics captured by the model. A zoom or zoom-out operation is applied to each sample so that the

target SCDs occupy 10–35% of the image area. The resolution of each input image is encoded into the outputs of the backbone using Feature-Wise Linear Modulation (FiLM) to provide scale context.

To detect SCDs at several scales, an image pyramid strategy is utilised at inference (Figure 1c). Predictions are generated at several scales by resampling the native DEM to a range of resolutions. The predictions from each resolution sweep are stitched together using a support-based voting system. For an object to become a final prediction, it must have been predicted in at least two scale sweeps at >65% confidence, or one sweep at >90% confidence. The final predicted objects are passed through a veto classifier, which either rejects or retains the prediction (Figure 1d). The classifier uses spectral imagery, wider DEM context, and DEM-derived scalars to reject predictions that exhibit characteristics inconsistent with typical SCDs, such as anthropogenic or unrelated natural features. The veto classifier uses two ResNet18 backbones to encode the spectral and topographic characteristics of the candidate SCD and the surrounding area. Scalar values representing the physical morphology of the candidate SCD (i.e., depth, absolute size) are encoded using a Feed-Forward Network (FFN).

2.2 Transfer Learning and Training

The Mask R-CNN is trained for 50 epochs using 590 samples at a range of resolutions. The backbone (ResNet50, FPN) is initialised with pretrained weights and incrementally unfrozen during training at very low learning rates to enable fine-tuning to the task while limiting overfitting to the small dataset. The shallow backbone layers are completely frozen during training to ensure the feature extractor does not overfit to the training dataset. A stepped learning rate scheme is applied, with different learning rates assigned to individual model blocks. This enables the AdamW optimizer to refine the model parameters and converge towards a minimum in the loss manifold. The veto classifier is trained for 50 epochs on false positives outputted by the Mask R-CNN. A weight decay of 1×10^{-4} is used for the optimisers of both models to aid generalisation.

The Mask R-CNN model is trained using Leave-One-Out Cross-Validation (LOOCV) to measure cross-geography generalisation on unseen regions. Model performance during both training and inference is evaluated using object-based F1, precision, and recall, as well as Intersection over Union (IoU) (Appendix A). Object-based metrics are employed over pixel-based metrics to inform performance in an exploration context, where object mask quality is less important than object identification. A true positive (TP) prediction must satisfy one of two criteria: (1) the centroid of the predicted SCD within a corresponding ground truth SCD mask, or (2) the predicted SCD mask and the ground truth SCD mask must have an $\text{IoU} \geq 0.5$.

2.3 Field Target Selection and Soil Gas Sampling

A localised, region-specific segmentation UNET was developed to assist manual SCD identification in Southern England before the cross-geography MaskRCNN was developed, due to scheduling constraints regarding fieldwork dates. The model was trained on 83 hand-picked depressions in the target region and used three input channels: digital elevation model, multidirectional hillshade, and topographic position index (TPI). The predictions of this model informed field target selection.

Soil gas sampling was performed using a Gembio portable gas analyser, capable of identifying H_2 , CO_2 , CH_4 , NO_2 , and O_2 . The analyser tube was placed into a 1-metre-deep drill hole within the soil. Drill holes were made using only percussion action to limit the risk of friction-related hydrogen artefacts, as first documented by (Halas et al., 2021). The temperature of the drill bit was regularly checked during sampling to ensure that no rotary friction occurred.

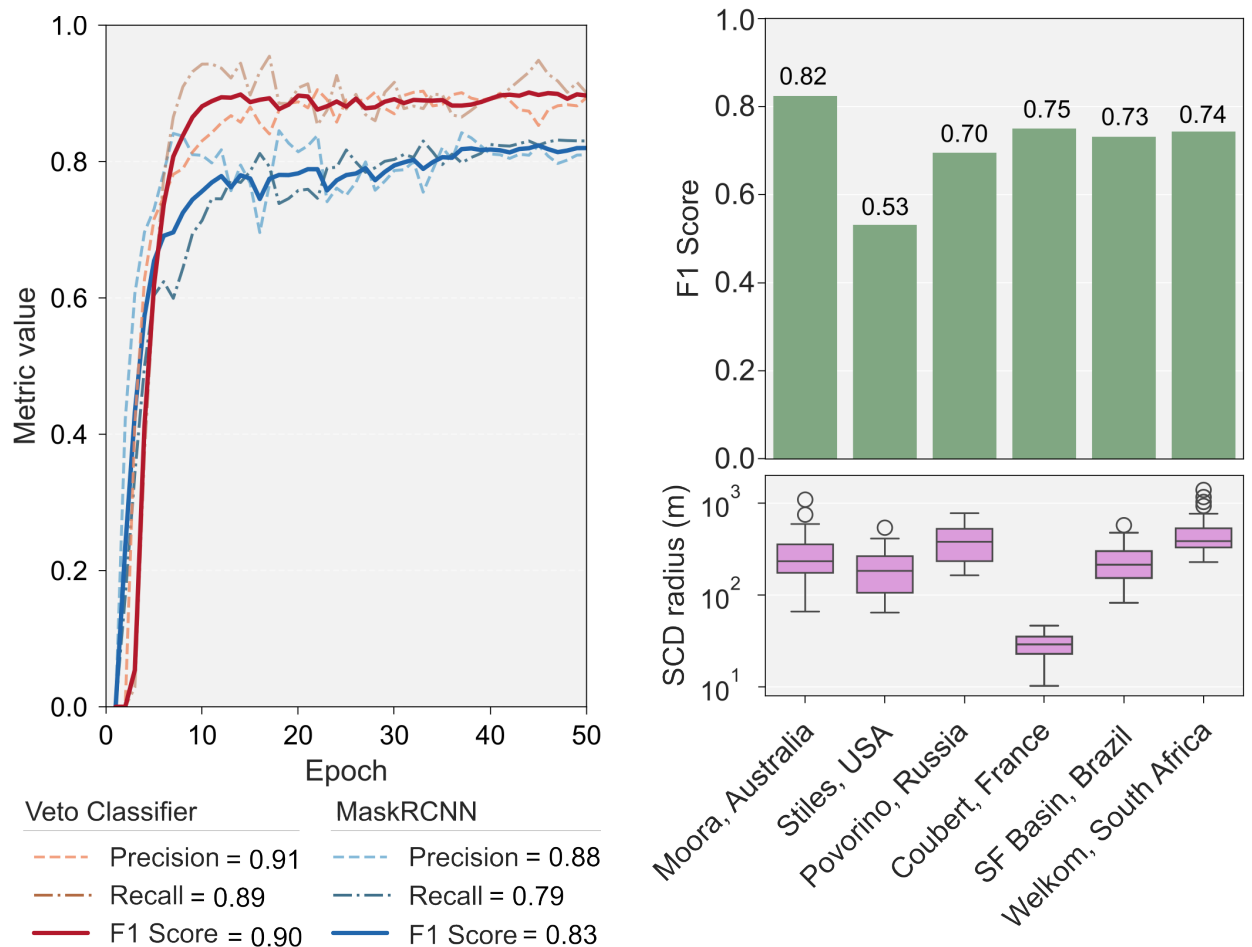


Soil gas measurements were taken at 15 SCDs. A separate 21-kilometre coastal transect was performed with 34 sample sites positioned approximately every 500 metres apart where

possible. The upper background threshold of H_2 concentration is calculated using the Median Absolute Deviation (MAD) of the average hydrogen concentrations sampled at each site to mitigate the effects of large positive outliers and the variability of hydrogen concentrations at each locality. The background upper threshold is defined as the Median + $(2 \times \text{MAD})$. A one-sided Kolmogorov-Smirnov D^+ statistic is used to analyse SCD enrichment near faults, with significance testing corroborated through Monte Carlo simulations.

3 Results

3.1 Model and Inference Strategy Performance



(a) Learning curves for the Mask R-CNN model and veto classifier. The stated metrics are from the best performing (highest F1 Score) epoch per model.

(b) SCD size distributions and cross-validation F1 scores per region for the Mask R-CNN. Model performs well in most regions despite a large range of physical SCD sizes.

Figure 2: Visualisation of Mask R-CNN and classifier training and validation metrics.

The Mask R-CNN successfully identifies SCDs in a random train-test split and achieves an F1 Score of 0.83, with a precision of 0.88 and recall of 0.79 (Fig. 2a). Crucially, the model also performs well in LOOCV, achieving an average F1 score of 0.71 in completely unseen regions which often contain different SCD morphologies and sizes, as well as general topographic differences (Fig. 2b). The scale-normalised LOOCV training and validation sets demonstrate that the model can correctly identify SCDs over a large physical scale range when the input samples are normalised to 512 by 512 pixels. For example, the model achieves an F1 score of 0.75 on the validation fold for Coubert, France, where the median SCD radius is 29 metres. In regions with much larger SCDs, such as Moora, Australia, where the median SCD radius is 233 metres, the model achieves an F1 score of 0.82. The veto classifier achieves an F1 Score of 0.90 in a random train-test split. For both models, the best model state (highest F1 epoch) is saved and utilised at inference to avoid overfitting.

The image-pyramid inference strategy successfully enables detection of SCDs at several scales. Importantly, deploying the model with an image pyramid enables automatic detection

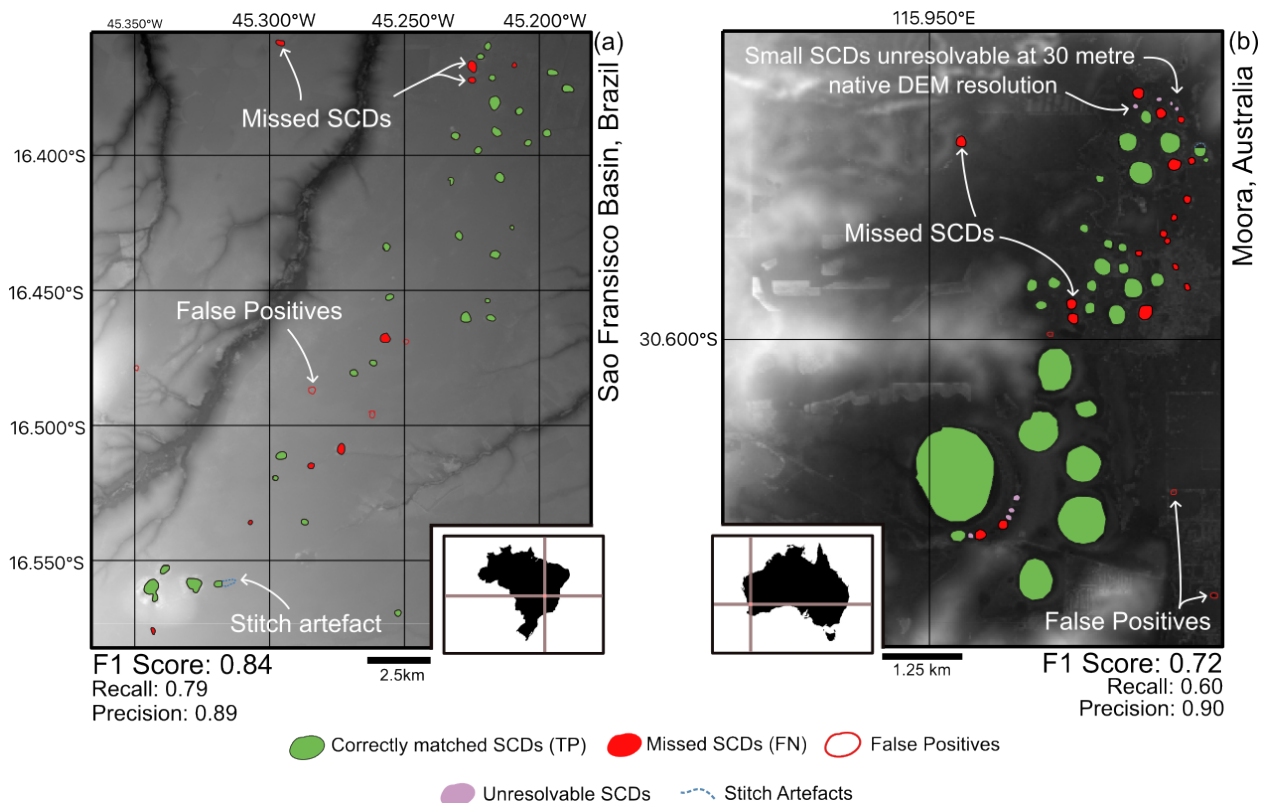


Figure 3: Inference predictions and object-based evaluation metrics: (a) predictions and ground truth comparison from the Sao Francisco Basin, Brazil, (b) predictions and ground truth comparison from Moora, Western Australia. Predictions are generated using an image pyramid with sweeps at 3, 6, 9, 12, and 15 metres per pixel.

of SCDs at any scale, without prior knowledge of the typical morphology of SCDs in the target region. Across different regions, the model correctly identified SCDs ranging from 25 metres to 1.5 kilometres in diameter. Inference predictions from the São Francisco Basin, Brazil, achieve an object-level F1 score of 0.84, with a precision of 90%. Visually, the model can accurately distinguish background from potential SCDs, but can struggle to confidently classify candidate structures, leading to lower recall and regular false positives. In Moora, Western Australia, the model consistently identifies large SCDs (500 metres – 1.5 kilometre diameter), but recall drops with SCD size due to resolution constraints. In regions where a 1 metre DEM is available (e.g., UK, France), small depressions (<100 metres) are consistently identified by the model. However, a consequence of the fine detail offered by these 1 metre DEMs is the overprediction of anthropogenic structures (such as farm ponds), as samples derived from fine DEMs are underrepresented in the training data due to data availability constraints. Coarser DEMs only reveal larger-scale topographic information and the model suffers less from overprediction problems, but they limit the lower size range of resolvable SCDs.

At inference, the low-threshold Mask R-CNN generates hundreds of predictions across multiple resolution sweeps. The veto classifier can remove up to 90% of the predictions if the region contains very few (or no) SCDs. Despite the majority of the removals being correctly identified false positives, the veto classifier occasionally removes true positives. Similarly, false positives occasionally pass through the veto classifier, especially when the candidate structure is small because the spectral resolution (3 metres per pixel) does not provide enough detail for a confident rejection. Artefacts from merging predictions from different resolutions occasionally affect the quality of the instance masks. Despite utilising a 25% tile overlap and Hann windows to reduce edge-effects, masks can occasionally be incomplete and blocky for less confident predictions (e.g., stitch artefact in Figure 3).

3.2 Model Inference and Cluster Identification in Southwest England

When deployed on 11,460 km² of Southwest England, the model predicts 2752 SCD candidates (Fig. 4). Visual inspection of the predictions in this region reveals that the Mask

R-CNN inference pipeline performs similarly well to the region-specific UNET developed for initial field target selection in the area. The Mask R-CNN successfully identifies SCDs at the correct size range (20 to 50 metres in diameter) in a completely unseen region, with no prior information about the size distribution of SCDs. Most predictions were made at the appropriate scale level (1 metre/pixel), with very few larger false positives predicted in coarser sweeps. However, due to the fine detail provided by the 1-metre DEM, the final predictions do contain anthropogenic false positives. The false positives are relatively evenly distributed and generally do not inhibit the capability of the model to identify SCD clusters (Fig. 4). Similarly, the model is not able to distinguish between SCDs based on origin; depressions associated with karst or bomb cratering and depressions associated with hydrogen leakage are predicted at the same confidence if they share similar morphologically characteristics (Fig. 4).

Despite lower precision at inference compared to Mask R-CNN validation, the model detects several clusters of depressions within the prospected region (Fig. 5). Two of the detected clusters are associated with sinkholes in documented karstic terranes, such as the Mendip Hills (Coleman and Balchin, 1959) and chalk dolines in Dorset (Sperling et al., 1977). Outside of these clusters, the primary enrichment in SCD density occurs in the central area, in proximity to the Watchett-Cothelstone-Hatch Fault (WHCF).

3.2.1 Field Results and Soil Gas H_2 Concentrations in the Bristol–Bray Cluster



Soil gas measurements in the Somerset area reveal elevated H_2 concentrations. Conversely, soil gas measurements from the field campaign in Cornwall are consistently lower, indicating less potential for active H_2 migration. The model detects 92 SCD candidates in the Somerset field campaign region, concentrated around two main clusters on the southwestern and northeastern side of the Quantocks Hills. Of the 15 depressions visited during fieldwork, 10 are successfully detected by the model. In the densest clusters, there are up to 1.6 SCD predictions per square kilometre.

Out of the 49 sample sites (34 along the transect, and 15 identified SCDs) in Somerset,

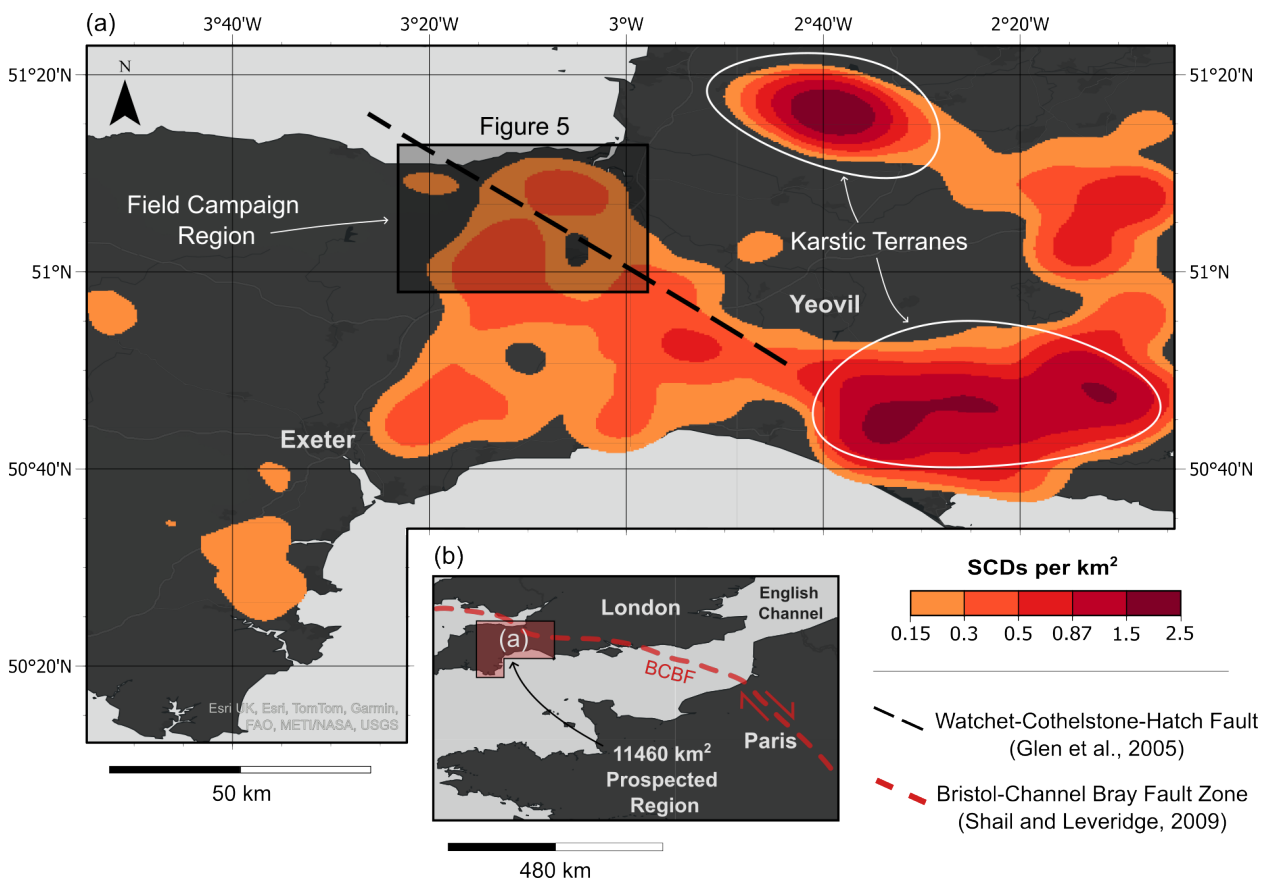


Figure 4: Density of SCD candidates predicted using the Mask R-CNN in the 11460 square kilometer prospected region of Southwest England. Predictions are generated using an image-pyramid with sweeps at 1, 3, 6, 9, 12, and 15 metres per pixel.

15 localities exhibit average H_2 concentrations exceeding the regional background upper threshold (9 ppm). A maximum concentration of 300 ppm is detected at Blue Anchor Beach. The median concentration of the 34 localities along the transect is 4.75 ppm. In total, the maximum concentrations at 9 localities measure above 40 ppm. Out of the 15 investigated SCDs, 7 exhibit mean soil gas concentrations above the background threshold, with four of these SCDs also exhibiting mean H_2 concentrations above 20 ppm. Five SCDs exhibit mean H_2 concentrations lower than 5 ppm. The SCDs in the region are small, with an average diameter of 27 metres and a depth of 1.8 metres. Due to the undulating topography of the region, the depressions are often not fully closed on the downslope side.



Localities associated with SCDs commonly contain higher soil gas H_2 concentrations compared to non-SCD localities, with mean concentrations of 11.6 and 7.2 ppm, respectively. However, the average hydrogen concentrations of SCD and non-SCD localities are not statistically separable into two populations (Mann–Whitney, $p = 0.247$). In comparison, the means of the peak H_2 concentrations measured at each SCD and non-SCD locality are 30.3 and 13.3, respectively. Using the maximum recorded values at each site, SCD and non-SCD localities are statistically separable (Mann–Whitney, $p = 0.0342$).

Hydrogen concentrations are highly variable both within and surrounding the SCDs (Figure 5b), demonstrating local heterogeneity in gas migration and accumulation within the soil. Depression rims generally exhibit higher H_2 concentrations than the depression cores, which commonly exhibit background values (approximately 0–5 ppm). These inactive cores bring down the average concentrations measured in SCDs; the median of the maximum concentrations measured at each SCD is 25 ppm. At every SCD, the highest H_2 concentration was detected either on the depression rim or 10–50 metres outside the structure.

There is no clear spatial correlation between elevated H_2 signals and the Watchett-Cothstone-Hatch Fault (WCHF), a major strike-slip Variscan fault. Similarly, there is no consistent relationship between visible coastal faults and increased hydrogen concentra-

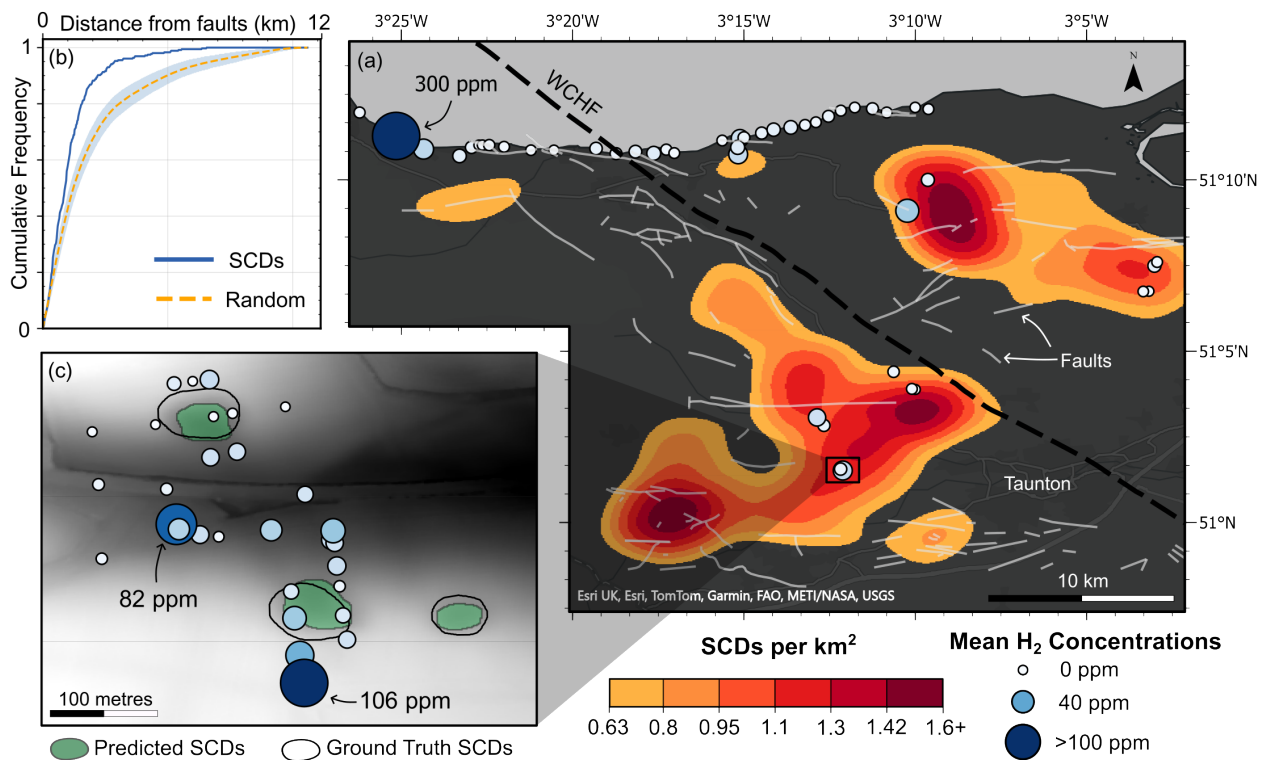


Figure 5: Model predictions and field campaign results in the Bristol–Bray Fault Zone, Somerset: (a) Soil gas H_2 measurements along a 21 kilometre coastal transect and 15 SCD localities, with model SCD prediction densities. Mapped faults (British Geological Survey, 1984) are indicated in white, (b) Distance distribution analysis demonstrating a spatial association between predicted SCDs and mapped faults, (c) Soil gas H_2 measurements at a small cluster of three predicted and visited SCDs near Oake Green. The cores of the depressions contain no H_2 anomalies, while the rims and outer sections exhibit up to 106 ppm.

tions. However, spatial analysis and Monte Carlo simulations demonstrate highly significant concentration of predicted SCDs near to mapped faults (KS $D^+ = 0.221$, KS $p = 1 \times 10^{-16}$, Monte Carlo $p = 0.0002$, 5000 simulations). A strong spatial correlation is retained when analysis is restricted to the Watchet-Cothelstone-Hatch Fault (KS $D^+ = 0.184$, KS $p = 1 \times 10^{-11}$, Monte Carlo $p = 0.0002$, 5000 simulations). Observed SCD enrichment compared to a random distribution is maximised at 6.3 kilometres from the WCHF, and 1.12 kilometres when all mapped faults are analysed.

4 Discussion

4.1 Model Performance

LOOCV results demonstrate that the Mask R-CNN model can generalise across geographies and scales, allowing it to be deployed in completely unseen regions. Similarly, inference results in out-of-fold regions similarly demonstrate robust geography and scale generalisation of the proposed model. Utilising a scale-normalised image-pyramid inherently enlarges the receptive field of the model, allowing identification of SCDs of different sizes. As discussed above, SCD sizes can differ by up to two orders of magnitude (Fig. 2b). Current models for SCD detection only utilise one resolution, limiting the range of detectable SCDs to that of the receptive field of the underlying neural network. Although these models are more tailored to a specific resolution, and may achieve better results on some regions, they are not applicable or generalisable to regions with smaller SCDs, such as the UK and France. Similarly, they are not applicable for unseen exploration in regions with SCDs that are not the same size as the depressions in the respective training dataset. As such, a scale-normalised image-pyramid inference approach is a crucial improvement over other current SCD detection strategies for first-pass exploration.

However, the model does not achieve full scale invariance, with smaller SCDs more commonly missed by the model despite the image-pyramid strategy. The failure mechanism here is clear; while spectral identification can utilise high-definition (3 metres per pixel) satellite imagery, topography-based identification relies on the global Copernicus 30 metre DEM unless a better national open-sources dataset is available. As a result, in regions where the native DEM resolution is 30 metres per pixel (e.g., Copernicus DEM), SCDs smaller than approximately 60 metres in diameter are unresolvable, as they comprise only a few pixels at the native resolution. This is primarily a data issue, and no image pyramid implementation can resolve resolution limitations of the source data. For example, in regions where a fine national DEM is available (e.g. 1 metre DEMs in the UK and France), small depressions are consistently identified by the model. While spectral-based models can achieve greater scale-invariance, the highly variable optical and spectral signatures of SCDs around the world limit the cross-geography generalisation. Although SCDs are often clearly identifiable through spectral anomalies and satellite imagery in arid regions (Moretti et al., 2022; Prinzhofer et al., 2019; Es-sabbar et al., 2026), in temperate, agricultural regions such as France, Russia and the UK, they display almost no consistent spectral or optical anomalies (Larin et al., 2015). Topography-based models do not suffer from a similar issue, as they morphological characteristics of SCDs are relatively similar across regions.

Inference results also suffer lower precision compared to Mask R-CNN validation due to large class imbalance between the normal background and rare SCDs in large deployment regions (several 1000s km²). False positives are more common with a 1 metre DEM due to increased anthropogenic features and fine-detail, despite heavy negative training with these resolutions. The model also predicts depressions that are not formed from H₂ migration, such as karstic sinkholes. However, these depressions can provide a preferential pathway for gas flow, and are still targets for model detection.

4.2 Hydrogen Prospectivity in Southwest England

Soil gas H₂ concentrations measured in this study provide the first promising indication of an active H₂ migration system in the United Kingdom. The maximum concentration of 300

ppm measured at Blue Anchor Beach exceeds the regional median (4.75 ppm) and typical background values (0–7 ppm; Li et al., 2009).

SCDs in the region are markedly smaller (20–50 metres diameter) than structures identified in other regions of active H_2 migration (e.g., Russian Craton; Larin et al., 2015), and often present a unique unclosed morphology due to undulating terrain. Despite morphological differences, the SCDs often exhibit the typical elevation of H_2 concentrations around the depression rim compared to the core. This is a characteristic trait of topographical structures formed from gas-driven expansion and subsequent collapse, leaving a low-permeability core and a ring of preferential flow pathways around the depression rim (Larin et al., 2015; Prinzhofer et al., 2019; Schopfer et al., 2025; Zgonnik et al., 2015). The presence of an ‘inactive’ subgroup of SCDs demonstrates that not all visited or predicted depressions are currently associated with H_2 migration, although some ‘inactive’ structures may originate from previous H_2 migration.

Statistically significant clustering of predicted SCDs near mapped faults strongly indicates fault-related gas migration. The governing WCHF demonstrates a similar statistical significance on the spatial distribution of SCDs, suggesting that it may exert a primary control on SCD formation. The WCHF inherently exhibits a much broader control (peak enrichment distance = 6.3 km) compared to the effect of all mapped faults (peak enrichment distance = 1.12 km). The broader control of the WCHF implies that it may act as a primary migration pathway, feeding smaller faults that distribute (or previously distributed) H_2 throughout the region. Although present-day H_2 signals do not correlate with these faults SCD clusters may have formed around previous migration pathways which are now inactive.

The source of H_2 in the region is unclear. The ultramafic Lizard Complex in Southwest England forms part of the Rhenohercynian Suture (Lefeuvre et al., 2024; Shail and Leveridge, 2009). Ultramafic rocks (serpentinite, peridotite) readily facilitate the production of H_2 through serpentinisation at the correct temperatures (200–300 °C) and silica conditions (McCollom et al., 2016). The oxidation of ferrous to ferric iron through reaction with water reduces the hydrogen atoms in the water to molecular H_2 (McCollom et al., 2016). The presence of ultramafic rock along the Rhenohercynian suture, which is bounded by the Bristol–Bray fault to the north and east (Shail and Leveridge, 2009), may indicate a possible ultramafic source rock at depth. Moreover, the major Variscan faults may extend down to the upper mantle and provide a direct hydrological link to basement ultramafics, facilitating large-scale H_2 production and migration. Magnetic and gravimetric data indicate the presence of granite in the north of the Paris Basin, indicating the possible role of radiolysis in H_2 generation (Lefeuvre et al., 2024). However, the exact source of H_2 in both the Paris Basin and Somerset is still unclear.

Although appreciably elevated, the measured anomalous concentrations (50–300 ppm) are moderate in the global context (e.g., thousands of ppm; Larin et al., 2015; Prinzhofer et al., 2019; Zgonnik et al., 2015). However, in the Paris Basin, on the southwestern side of the Bristol–Bray Fault System, only 9 out of 240 measurements exceed 100 ppm, despite the underlying aquifer containing up to 52 volume percent H_2 (Lefeuvre et al., 2024; Léger et al., 2025). As such, the measured H_2 concentrations and significant fault-related SCD clustering demonstrate substantial potential for an active subsurface H_2 migration system. Further isotopic investigation of collected H_2 gas is needed to confirm the link between H_2 discoveries in the Paris Basin and Somerset. Similarly, geophysical investigation (i.e. seismic, gravimetric, magnetic) into the subsurface is required to understand migration pathways and identify potential granitic or ultramafic source rocks.

5 Conclusion

The performance of the model in validation and inference confirms that the Mask R-CNN image-pyramid pipeline provides a flexible, cross-geography tool applicable for automatic SCD detection. Similarly, utilisation of topographic data removes the limitation associated with the inconsistent spectral characteristics of SCDs. These contributions provide a crucial improvement to first pass natural hydrogen exploration, and specifically enable the first automatic identification of SCDs in the United Kingdom. Several clusters of depressions

are identified near the Bristol Channel–Bray Fault Zone. Significant enrichment of SCDs to mapped faults in the area, specifically the major Variscan Watchet-Cothelstone-Hatch Fault, indicate localised H_2 migration throughout the fault network. Soil gas concentrations of up to 300 ppm further demonstrate the promising potential of the region for further H_2 exploration, especially considering the geological link with highly documented discoveries in the Paris Basin.

6 Acknowledgements

To be written...

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